

DAY OF WEEK

```
class Solution {
    public String dayOfTheWeek(int day, int month, int year) {
        String[] daY =
{"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
        int[] daysOfMonth = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
        if( leapYear(year) ){
            daysOfMonth[2] = 29; // Feb has 29 days if it is a leap year
        }
        int daysOfYear = 4; //31st December 1970 is Thursday, so intialize as
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        //Add the days of the previous years
        for( int i = 1971 ; i < year ; i++){
            if( leapYear(i)){
                daysOfYear += 366;
            }else{
                daysOfYear += 365;
            }
        }
        //Add the days of the previous months
        for( int i = 1 ; i < month ; i++){
            daysOfYear += daysOfMonth[i];
        }
        daysOfYear += day; //Add the current day
        return daY[daysOfYear % 7]; //Modulo divide by 7 to find odd days
    }
    private boolean leapYear(int year){
        if((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
            return true;
        return false;
    }
}
```